Session 1
Introduction to energy

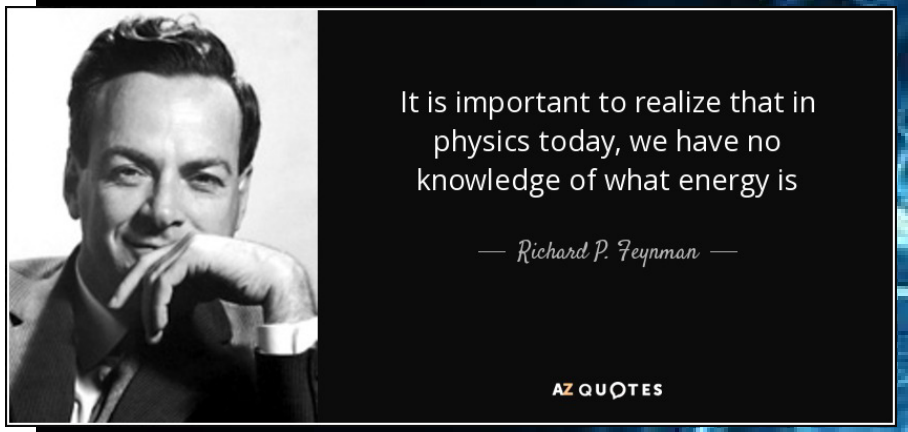
Jens Oluf Jensen

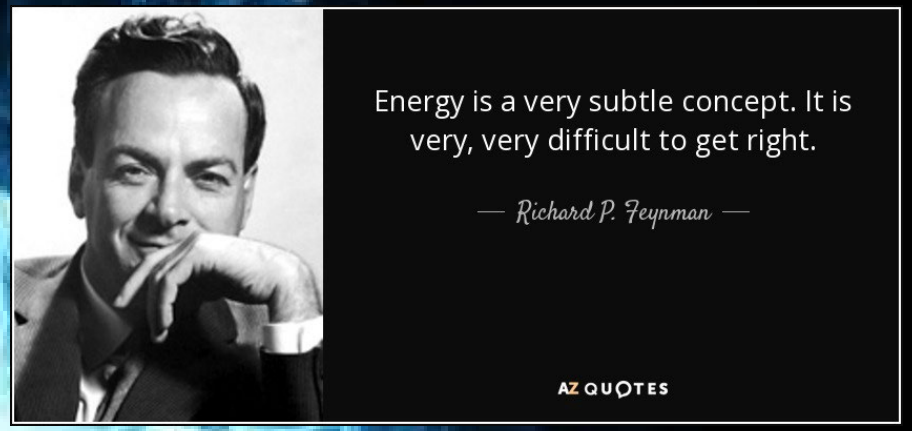
Department of Energy Conversion and Storage
Fysikvej, building 310
Technical University of Denmark
2800 Lyngby
Denmark
jojen@dtu.dk

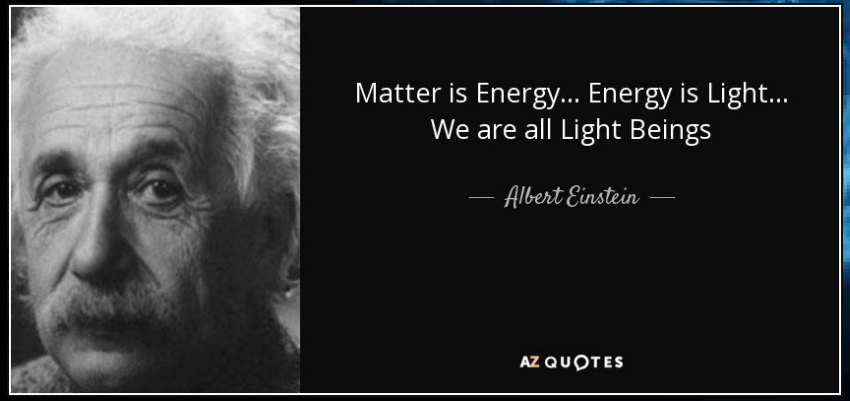
Energy: a widely accepted currency in the universe

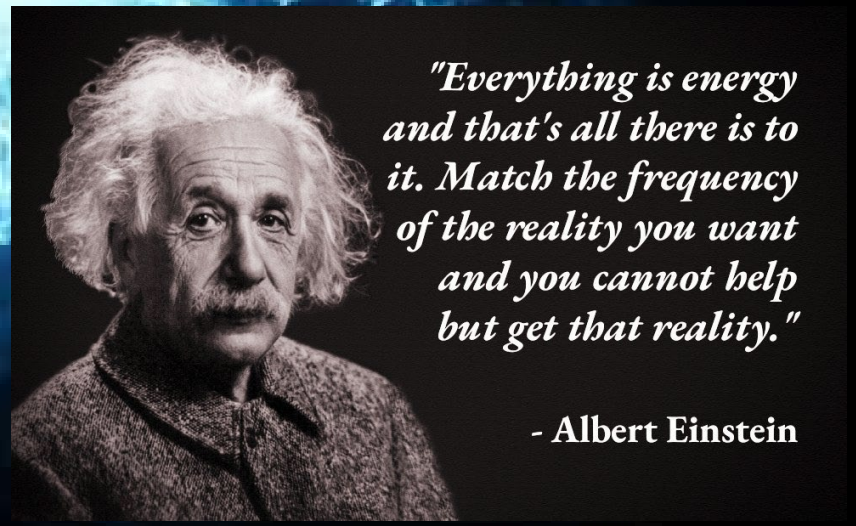


What is energy?









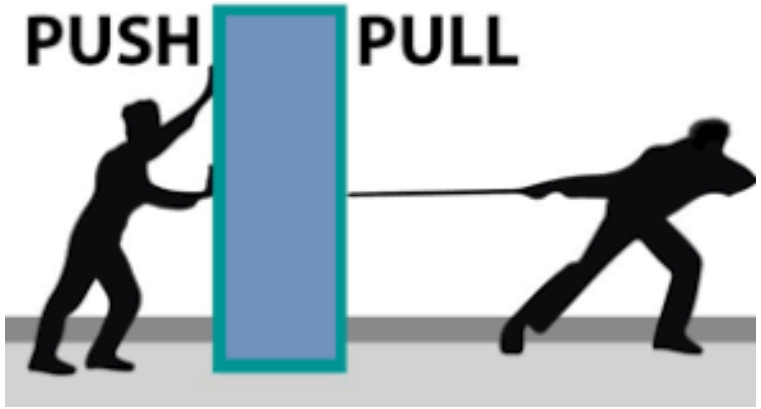
First law of thermodynamics

Energy is constant

- Energy cannot be created
- Energy cannot be destroyed

Energy, Power and force

Force and energy



Forces act on an object.

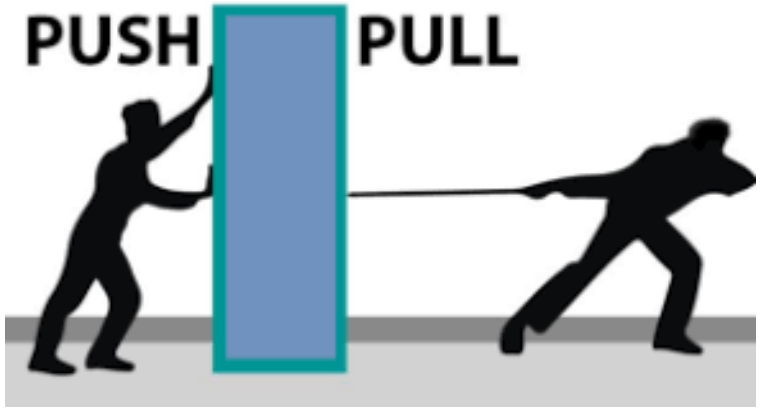
It is only when the object moves that energy is exchanged



Only the resulting force moves the object.

The energy transferred by a force is ***force times displacement*** $W = Fx$

Force and energy



The energy transferred by a force is *force times displacement*

$$W = Fx$$

More generally (if the force changes with displacement)

$$\mathrm{d}W = F\mathrm{d}x$$

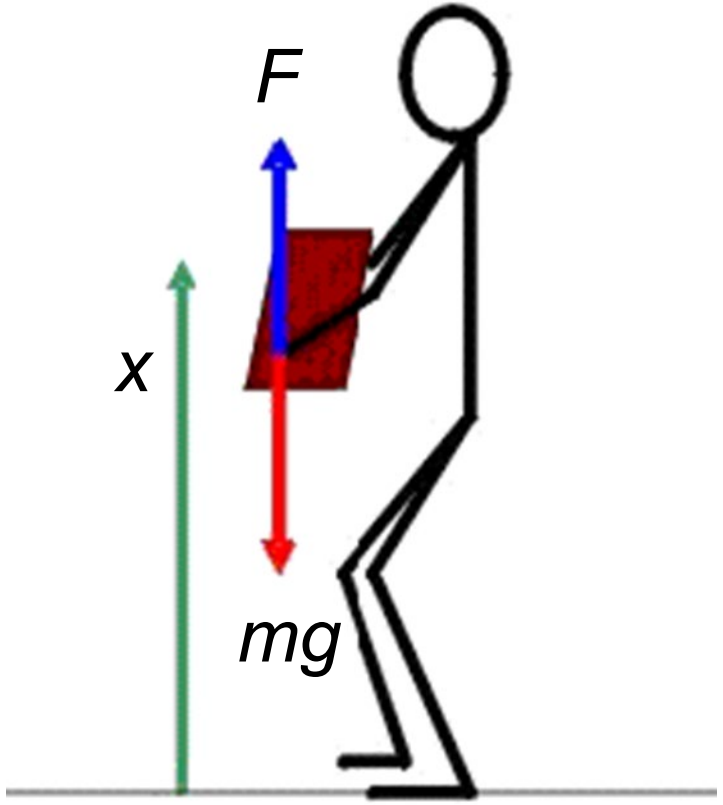
$$W = \int_{x_1}^{x_2} F\,\mathrm{d}x$$

Force energy and work

Work, *W*, is ordered movement against an opposing force

$$W = Fx$$





A process performs work if it can (in principle) lift a weight

Force, energy and work

Energy is energy – contained or transferred

A force may transfer energy – if displacement happened in the direction of the force

Power is a flow of energy – energy per time

Power is energy per time

Unit: watt, W = J s^{-1}

Watt	
Power	**Device**
1 W	Cell phone
10 W	Electric toothbrush
100 W	LED TV set

Kilowatt	
Power	**Device**
1 kW	Flatiron/dishwasher
10 kW	Car on highway
100 kW	Farmer’s tractor

Megawatt	
Power	**Device**
1 MW	Formula 1 race car/Ferry
10 MW	Container ship
100 MW	Aircraft carrier

Gigawatt	
Power	**Device**
1 GW	Power plant
10 GW	Large hydropower plant
100 GW	Japan’s electricity consumption
1 TW	China’s electricity consumption

Forms of energy

Kinetic energy

Solar energy

Mechanical energy

Tidal energy

Kinetic energy

Thermal energy

Electrical energy

Compression energy

Chemical energy

Radiation energy

Potential energy

Geothermal energy

Surface energy

nuclear energy

Wave energy

Magnetic energy

Rotational energy

Four fundamental forms of energy

Kinetic Energy	Gravitational Energy	Electrical energy	Nuclear energy
Wind	Hydropower	Electrical power	Nuclear reactions
Ocean currents	Tides	Capacitors	
Thermal motion (heat)	Waves	Chemical reactions	
Particle radiation	…	Electromagnetic radiation	
…		…	

Kinetic energy

Newton's laws

"*An object at rest stays at rest, and an object in motion stays in motion with the same speed and in the same direction unless acted upon by an unbalanced force*" (Newton's first law).

"*The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass*" (Newton's second law).

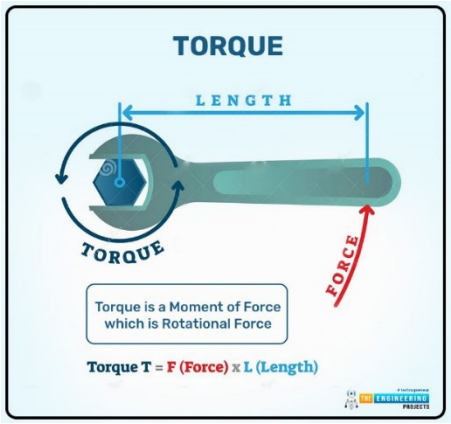
$$F = ma$$

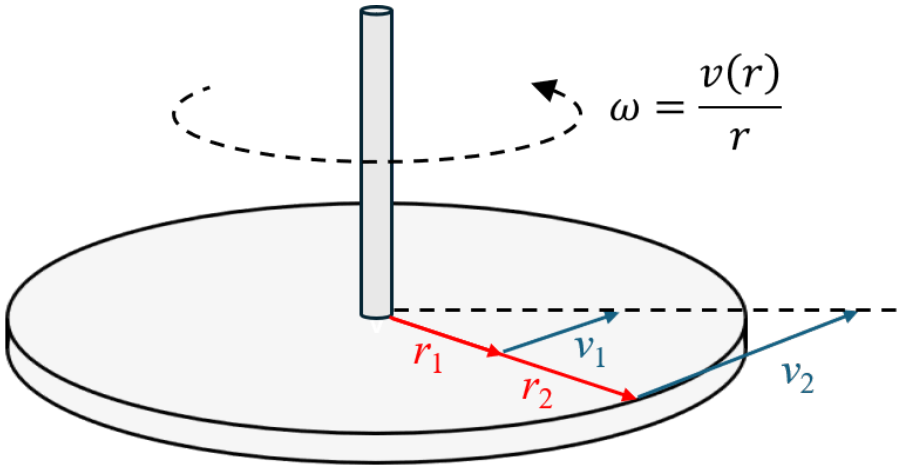
"*Action and Reaction. For every action force, there is an equal and opposite reaction force"* (Newton's third law).

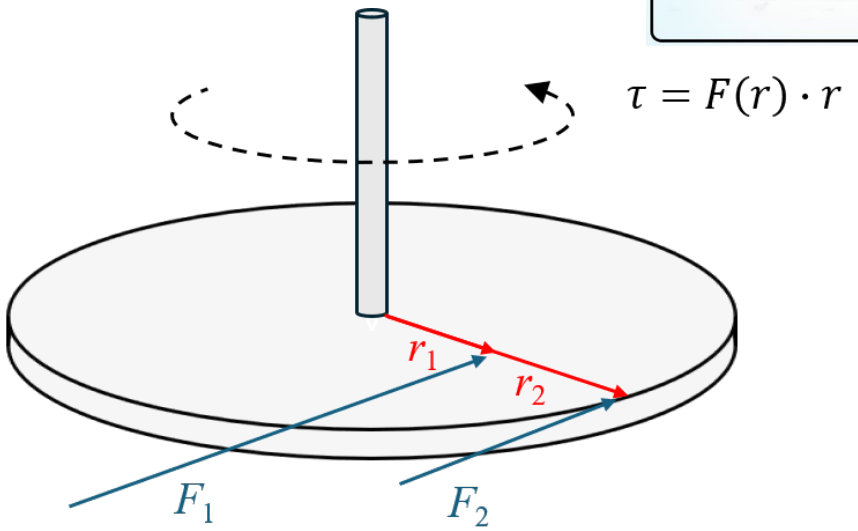
Kinetic energy

	Translational movement	Rotational movement
Force / Torque	$F = ma$	$\tau = I\alpha$
Energy	$E_{\text{kin}} = \frac{1}{2}mv^2$	$E_{\text{kin}} = \frac{1}{2}I\omega^2$
Energy gain	$\Delta E_{\text{kin}} = F\Delta x$	$\Delta E_{\text{kin}} = \tau\Delta(\text{angle})$

Rotational kinetic energy





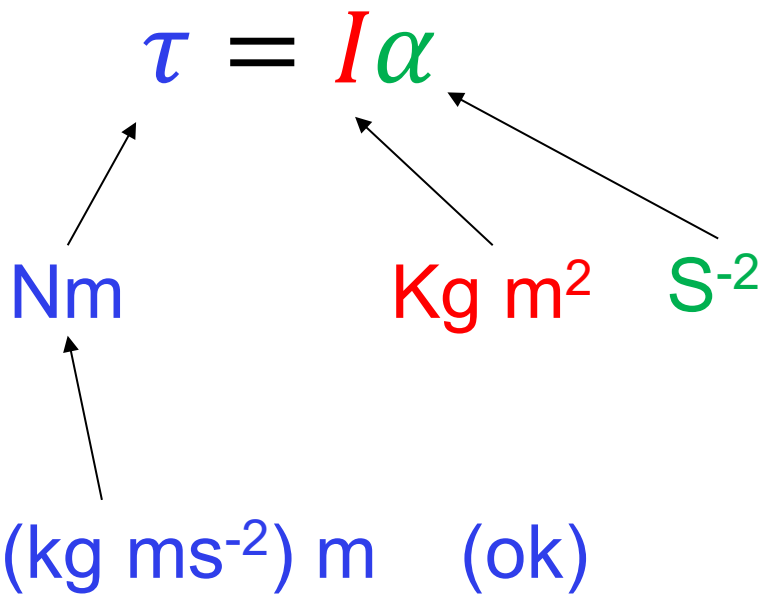


Translational speed is replaced by angular velocity, ω (rad s^{-1}).
Each mass element has kinetic energy, $E_{\text{kin}} = \frac{1}{2}mv^2$
Volume integration of all mass elements gives

$$E_{\text{kin}} = \frac{1}{2}I\omega^2$$

Linear force is replaced by torque, τ (Nm).
$F = ma$ is replaced by $\tau = I\alpha$,
where I (kg m^{-2}) is the moment of inertia and
α is angular acceleration (rad s^{-2})
$\Delta E_{\text{kin}} = F\Delta x$ is replaced by $\Delta E_{\text{kin}} = \tau\Delta(\text{angle})$.

Dimensional analysis



Gravitational energy

Mass attraction $$F = G\frac{m_1 m_2}{r^2}$$

The gravitational force is a function of distance.
However, on earth with + 6000 km to the centre of gravity, this can normally be neglected
You may use 9.81 m s^{-2} as a ”universal” acceleration of gravity on earth

$$E_{\text{pot}} = mgh$$

$$\Delta E_{\text{pot}} = mg\Delta h$$

F: force
G: gravitational constant,
$G = 6.6743 \cdot 10^{-11}\ m^3\ kg^{-1}\ s^{-2}$.
m: masses of bodies
r: distance
g: acceleration of gravity
E_{pot}: potential energy
h: height

Electrical energy

Electrical attraction/repulsion

$$F = k_e \frac{q_1 q_2}{r^2}$$

F: force (attractive or repulsive)

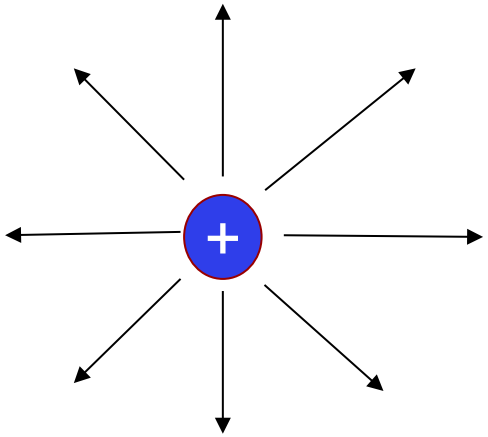
k_e: Coulomb constant,

$k_e = 8.9875 \cdot 10^9$ N m^2 C^{-2}.

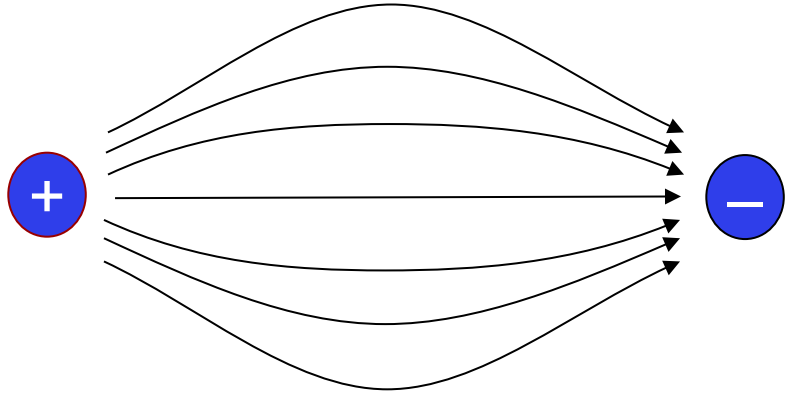
q: charges

r: distance

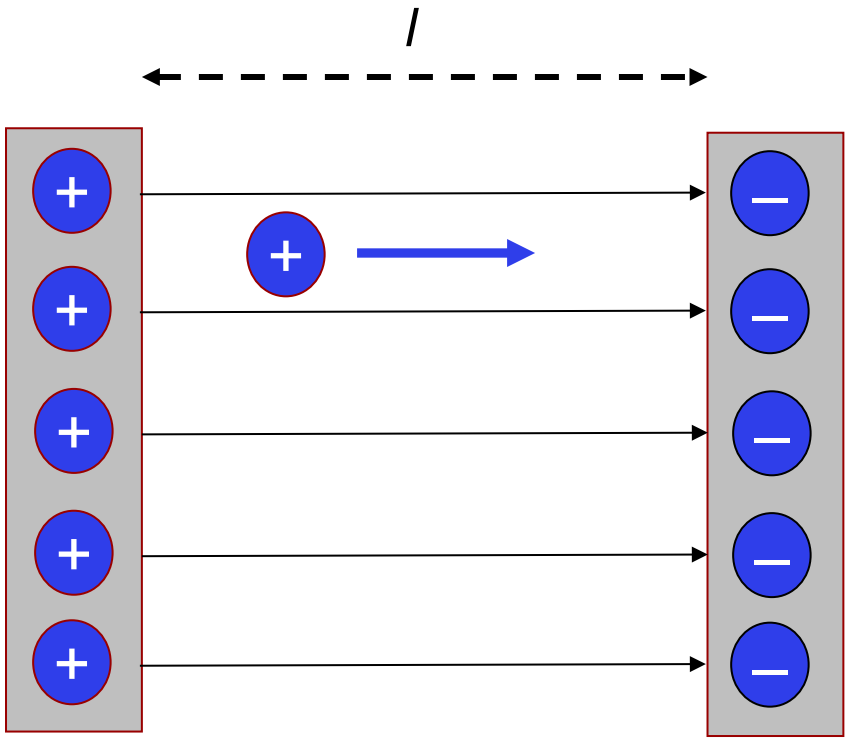
Field lines



Field lines



Electrical energy



Voltage difference (volt, V)

Field strength (volt per meter, Vm^{-1})

$$\Delta E_{el} = q\Delta U$$

$$F = k_e \frac{q_1 q_2}{r^2}$$

F: force (attractive or repulsive)

k_e: Coulomb constant,

$k_e = 8.9875 \cdot 10^9$ N m^2 C^{-2}.

q: charges

r: distance

l: distance

U: voltage difference

g: acceleration of gravity

E_{el}: electrical energy

The field can be through an electric wire

Electrical energy

Charge displaced in an electric field: $\Delta E_{el} = q\Delta U$

Power from voltage and current: $P = IU$

Electrical energy from power: $E_{el} = Pt$

Ohm's law: $U = RI$

Power loss in a wire: $P = RI^2$

Chemical energy (fundamentally electrical)

Braking of bonds requires energy

Formation of bonds liberates energy

$$CH_4 + 2O_2 \rightleftarrows CO_2 + 2H_2O$$

Hypothetical sub-reactions (breaking all bonds)

Energy1 + $CH_4 \rightarrow C + H + H + H + H$

Energy2 + $O_2 + O_2 \rightarrow O + O + O + O$

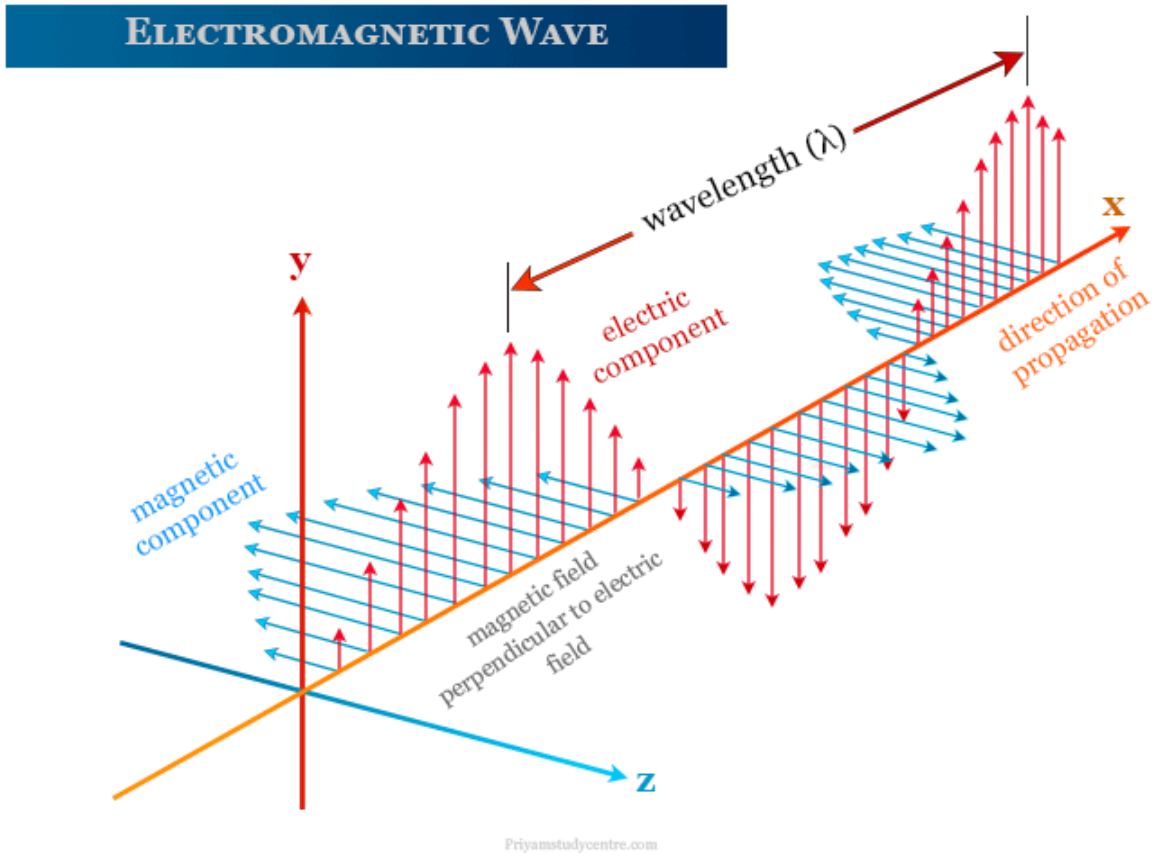
Forming new bonds

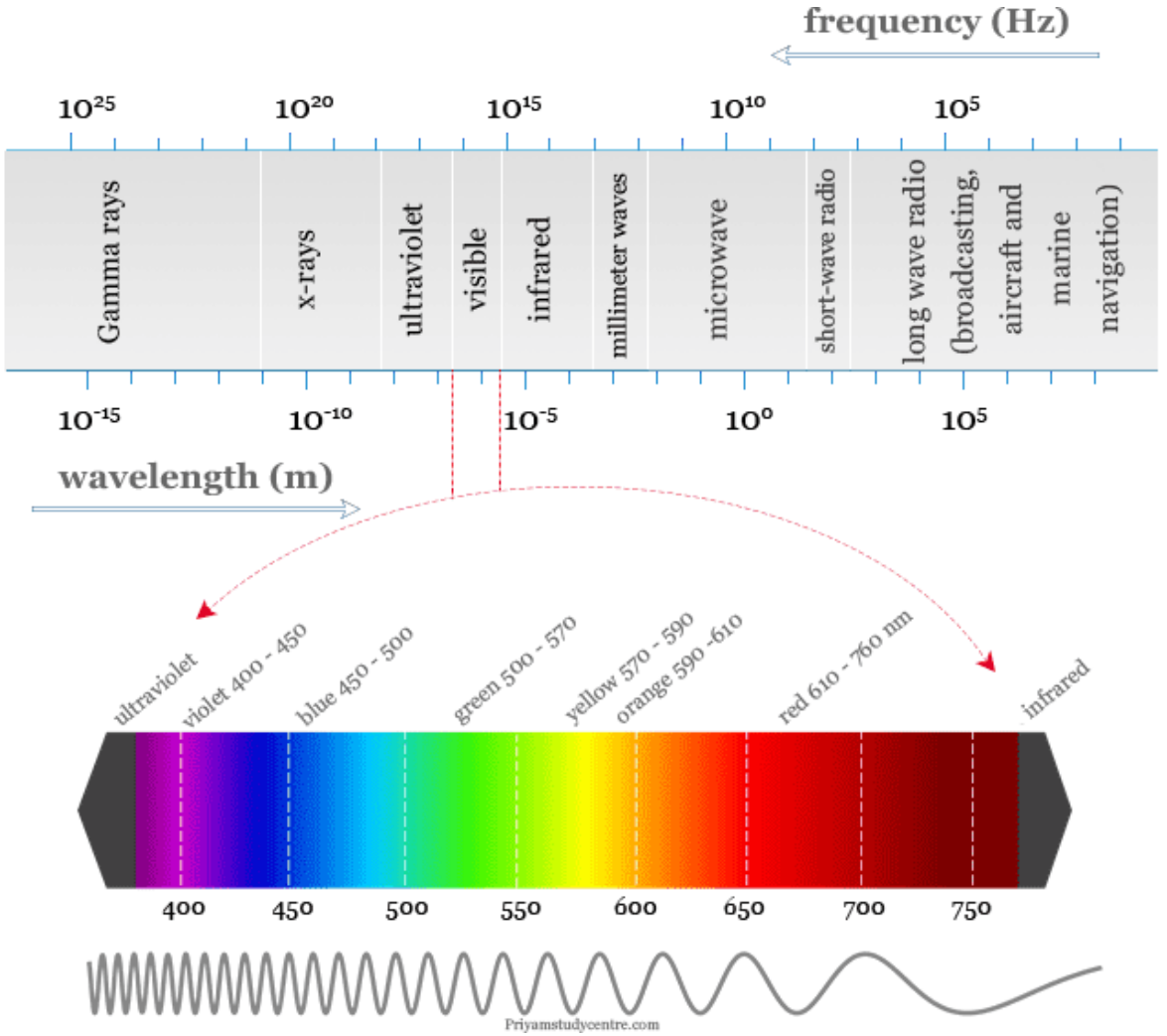
$C + O + O \rightarrow CO_2$ + Energy3

$H + H + H + H + O + O \rightarrow H_2O + H_2O$ + Energy4

Energy released (generated minus consumed) = Energy3 + Energy4 – (Energy1 + Energy2)

Electromagnetic radiation





from priyamstudycentre.com

Nuclear energy

Related to change in the atomic nucleus

Mass and energy are equivalent (same thing)

$$E = mc^2$$

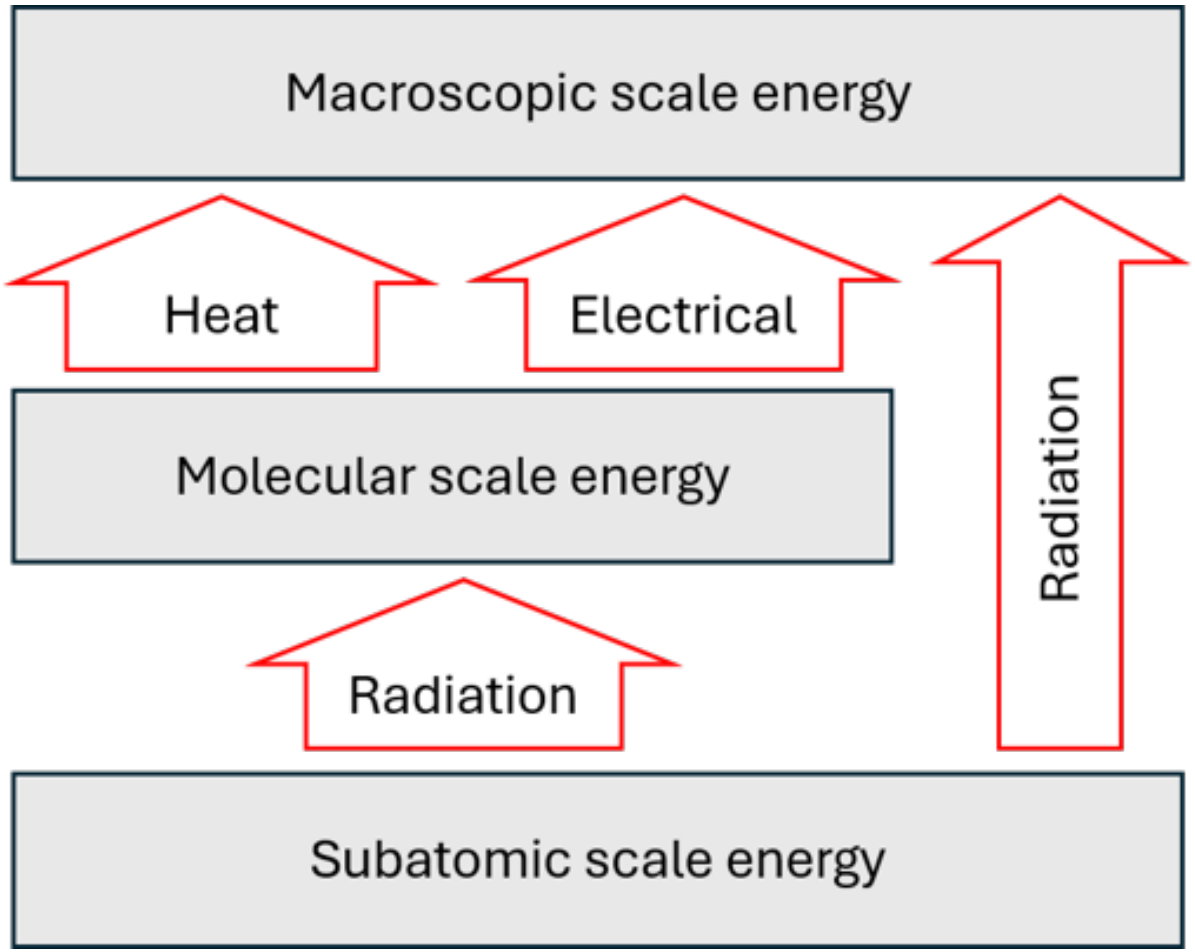
E: energy
m: mass
c: speed of light

Treated in separate lecture

Three fundamental energy regimes

	Kinetic Energy		Gravitational Energy	Electrical energy	Nuclear energy
Macroscopic scale	Wind	Heat	Hydropower Tides	Electrical power Capacitors	*Not relevant*
Molecular scale	Thermal motion		*Not relevant*	Chemical reactions	*Not relevant*
Subatomic scale	Particle radiation		*Not relevant*	Electromagnetic radiation	Nuclear reactions

Three fundamental energy regimes



SI units

Base units

Quantity	Name (SI unit)	Symbol
Time	second	s
Length	metre	m
Mass	kilogram	kg
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

Quantity	Name (SI unit)	Symbol	Intuitively	From base units
Energy	joule	J	N·m	$kg \cdot m^2 \cdot s^{-2}$
Power	watt	W	V·A	$kg \cdot m^2 \cdot s^{-3}$
Force	newton	N		$kg \cdot m \cdot s^{-2}$
Pressure	pascal	Pa	$N \cdot m^{-2}$	$kg \cdot m^{-1} \cdot s^{-2}$
Electric potential	volt	V	$J \cdot C^{-1}$	$Kg \cdot m^2 \cdot s^{-3} \cdot A^{-1}$
Charge	coulomb	C		s·A
Capacitance	farad	F	$C \cdot V^{-1}$	$kg^{-1} \cdot m^{-2} \cdot s^4 \cdot A^2$
Resistance	ohm	Ω	$V \cdot A^{-1}$	$kg \cdot m^2 \cdot s^{-3} \cdot A^{-2}$
Conductivity	siemens	S	Ω^{-1}	$kg^{-1} \cdot m^{-2} \cdot s^3 \cdot A^2$

A good idea to check calculations by a unit analysis. Are the units the same on both sides of the equation?

Capacity factor, CF

A solar panel does not produce at 100% all the time

CF is the quantification of the utilization of a device

$$CF = \int \frac{\text{Actual power}}{\text{Rated power}} \mathrm{d}T$$

$$CF = \frac{\text{Actual energy produced}}{\text{Rated power} \cdot \text{time}}$$

The time period can be any - a year, a lifetime, since start …

Higher and lower heating value

$CH_4(g) + 2O_2(g) \rightleftarrows CO_2(g) + 2H_2O(g)$ + energy (LHV)

(g) gas, (l) liquid, (s) solid

$CH_4(g) + 2O_2(g) \rightleftarrows CO_2(g) + 2H_2O(l)$ + energy (HHV)

Fuel	State at ambient temperature and pressure	HHV (MJ/kg)	LHV (MJ/kg)
Hydrogen	Gas	141.9	119.9
Methane	Gas	55.5	50
Ethane	Gas	51.9	47.8
Gasoline	Liquid	47.5	44.5
Diesel	Liquid	44.8	42.5
Methanol	Liquid	20	18.1

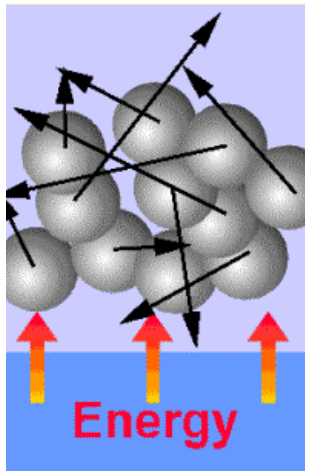
Expansion of a gas

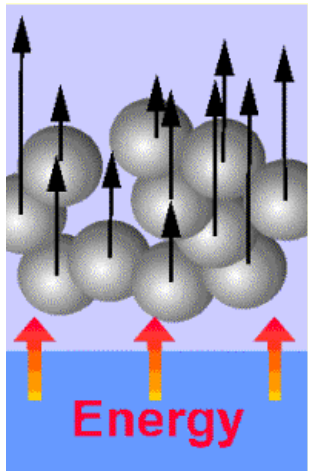
Ordered and disordered energy – work and heat

Heat

Disordered thermal energy transferred to/from the system.

Atoms/molecules are accelerated randomly



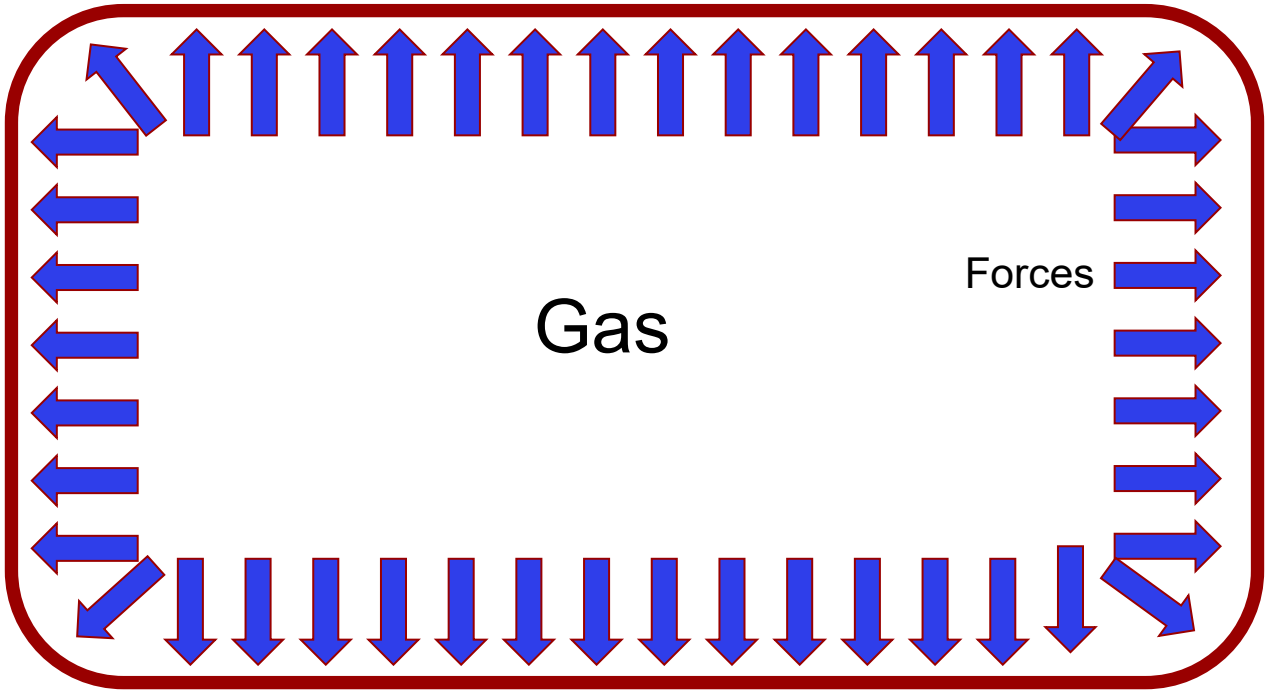


Work

Ordered mechanical (electrical) energy transferred to/from the system.

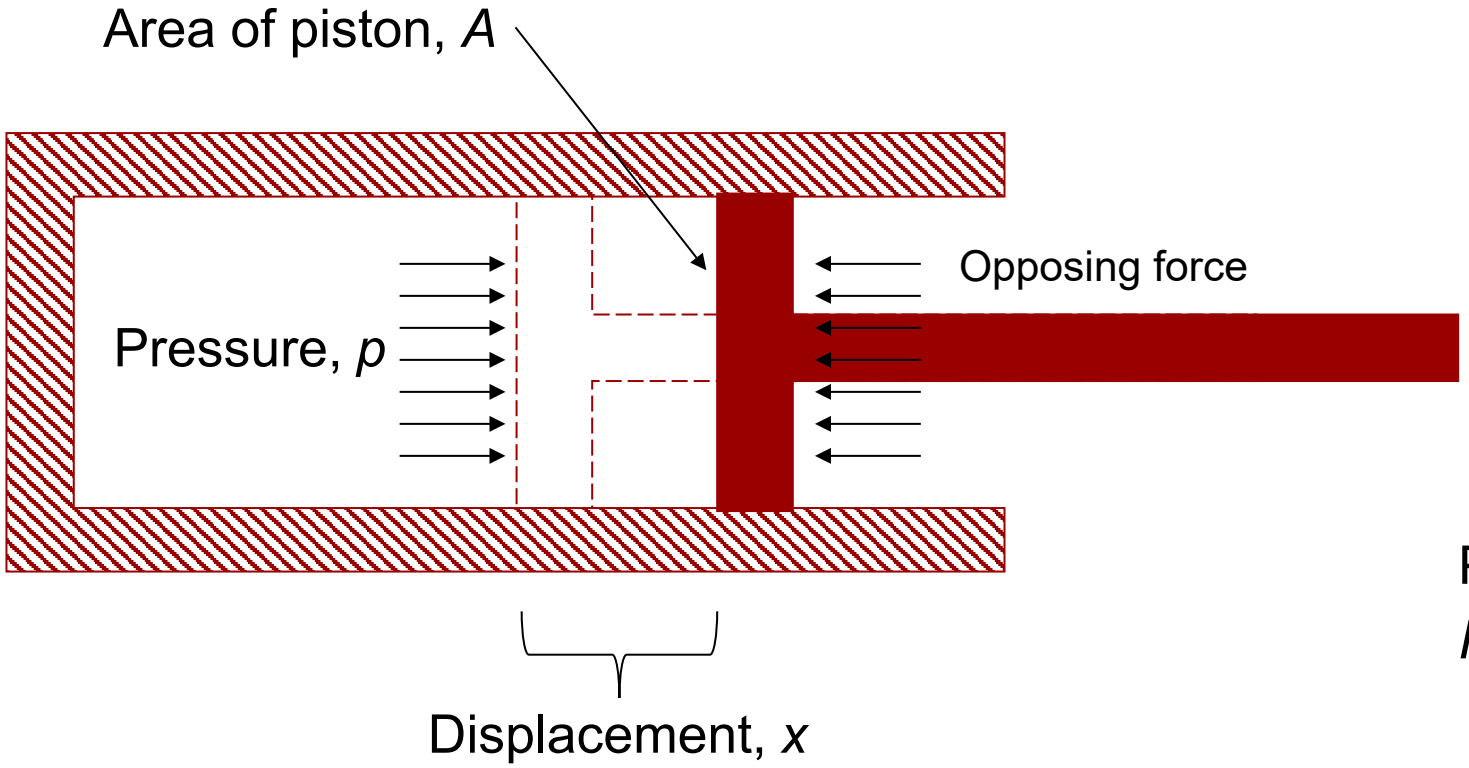
Atoms/molecules are accelerated ordered

Pressure



Pressure = force per area (N m^{-2})

Pressure-volume work (*missing in text*)



$W = F \bullet x$

Pressure = Force / area = F / A

$F = p \bullet A$

$W = F \bullet x = p \bullet A \bullet x = p \bullet V$

$$W = p \bullet \Delta V$$

Ideal gas – perfect gas

The ideal gas equation of state

Assumptions

- No interaction between molecules, except elastic collisions
- The volume of each molecule is negligiblev

$$pV = nRT$$

Validity

- Most accurate at low pressure and high temperature
- Breaks down at states near to condensation

Molar form $$pv = RT$$

p: pressure (Pa)
V: volume (m^3)
v: molar volume (m^3 mol^{-1})
n: no. of moles
R: the gas constant (8.314 J mol^{-1} K^{-1}
T: absolute temperature (K)

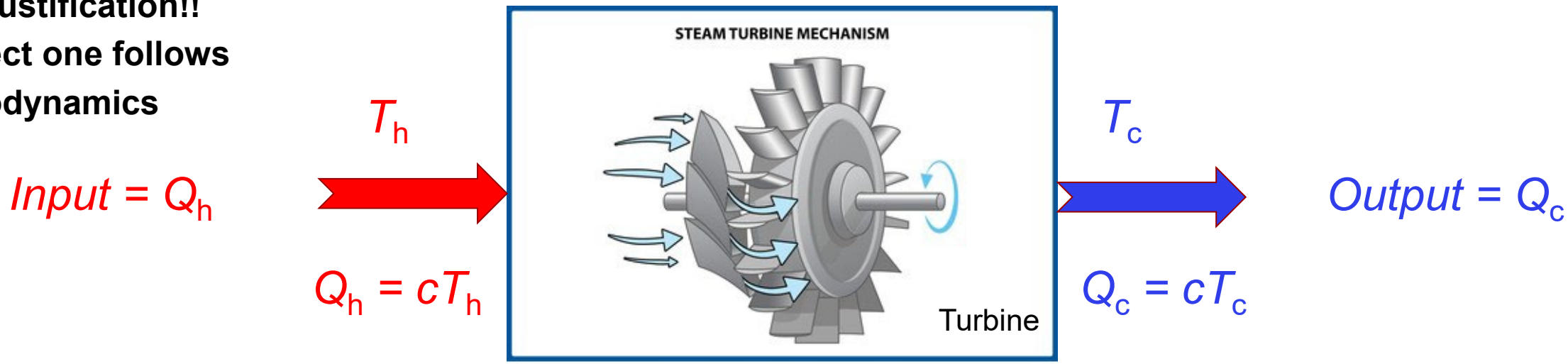
Energy efficiency

$$\text{Efficiency} = \frac{\text{What you get}}{\text{What you pay}}$$

Carnot's limitation (Carnot's law)

$$\eta_{\text{carnot}} = \frac{T_\text{h} - T_\text{c}}{T_\text{h}} \quad \text{or} \quad \eta_{\text{carnot}} = 1 - \frac{T_\text{c}}{T_\text{h}} \qquad W_{\max} = \eta_{\text{carnot}} Q$$

Intuitive justification!!
The correct one follows in thermodynamics



Work: $W = Q_\text{h} - Q_\text{c}$

$$\text{Efficiency} = \frac{\text{What you get}}{\text{What you pay}} = \frac{Q_\text{h} - Q_\text{c}}{Q_\text{h}} = \frac{T_\text{h} - T_\text{c}}{T_\text{h}}$$

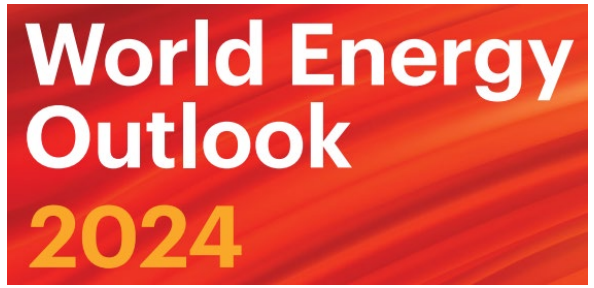
Sources of information on energy

Sources of energy statistics

International Energy Agency

Annually: World energy outlook

https://www.iea.org/reports/world-energy-outlook-2024



Energy statistics by country. Report: Statistical Review of World Energy

https://www.energyinst.org/statistical-review

Sources of energy statistics

US Department of Energy (EIA) International Energy Outlook

Not a report but links to numerous files with tables

https://www.eia.gov/outlooks/ieo/

BP, Statistical Review of World Energy (annual)

https://www.bp.com/en/global/corporate/energy-economics.html

Sources of energy statistics

Our World in Data

Lots of statistic and nice graphic presentations

https://ourworldindata.org

https://ourworldindata.org/search?q=Energy

Broad sources of energy information

International Energy Agency

Many energy reports

https://www.iea.org/

IRENA
International Renewable Energy Agency

Many energy reports

https://www.irena.org/